Fourier Neural Operators for 2D Navier–Stokes: A Comparison with U-Net

Ian Havill
havill@oxy.edu

[Group Members]
Occidental College
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Abstract

We compare the Fourier Neural Operator (FNO) [1] against a standard U-Net [2] on the task of predicting the one-step time evolution of 2D incompressible Navier–Stokes equations. Both models are trained on 10,000 vorticity field pairs at 128×128 resolution drawn from the neuraloperator Zenodo dataset [3]. FNO achieves a test relative-L^2 error of 0.0348 with 4.7M parameters, outperforming U-Net (relative-$L^2 = 0.1029$, 31M parameters) by a factor of roughly 3× while using 6.6× fewer parameters and running 2.2× faster. In zero-shot super-resolution experiments—where both models are evaluated at resolutions never seen during training—FNO's error remains flat (0.019 across 64, 128, and 256 grid points per side), while U-Net's error degrades by 4–6× at unseen resolutions. These results illustrate how architectures that encode physical structure in the frequency domain can outperform generic convolutional baselines on PDE surrogate tasks.

1 Introduction

Simulating fluid dynamics is a central problem in science and engineering, with applications ranging from weather forecasting and aircraft design to biomedical flow modelling. The governing equations—the Navier–Stokes (NS) equations—are well understood mathematically, but numerically solving them is expensive: a typical high-fidelity simulation on a fine spatial grid requires minutes to hours on powerful hardware, and must be repeated from scratch for each new set of initial conditions.

A natural alternative is to *learn* the simulation from data. Rather than solving the PDE numerically each time, one trains a neural network to approximate the solution operator $\mathcal{G} : u(t) \mapsto u(t+\Delta t)$ directly, and then evaluates it at inference time. If this operator can be learned efficiently, inference is a single forward pass taking milliseconds.

The key challenge is that the solution operator maps *functions* (vorticity fields) to functions, not fixed-length vectors. Standard convolutional networks (CNNs) are tied to the resolution they were trained at and carry no principled notion of function-space generalization. Li et al. [1] introduced the *Fourier Neural Operator* (FNO), which operates in frequency space and is, by design, resolution-independent.

In this work we benchmark FNO against a U-Net [2]—a widely used CNN architecture—on the 2D Kolmogorov-forced Navier–Stokes problem. We evaluate both models on (1) standard accuracy and inference speed, and (2) zero-shot super-resolution: the ability to generalize to grid resolutions the model was never trained on.

2 Background

2.1 2D Navier–Stokes Equations

We consider the 2D incompressible Navier–Stokes equations in vorticity form on the unit torus

$[0,1]^2$:

$$\frac{\partial \omega}{\partial t} + (\mathbf{u} \cdot \nabla)\omega = \nu \nabla^2 \omega + f, \tag{1}$$

$$\omega = \nabla \times \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0. \tag{2}$$

Here ω is the vorticity scalar, $\mathbf{u}$ is the velocity field, ν is kinematic viscosity, and f is an external body force. We use the Kolmogorov forcing $f(x, y) = \sin(4\pi y)$, which injects energy at low wavenumbers and drives the system into a turbulent regime. With $\nu = 10^{-4}$ the Reynolds number is $Re = 10{,}000$, placing the flow firmly in the turbulent regime.

The task is to learn the discrete time-stepping operator: given ω at time t, predict ω at time $t + \Delta t$.

2.2 Fourier Neural Operator

A classical integral operator kernel $\mathcal{K}$ applied to a function v can be written as a convolution:

$$(\mathcal{K}v)(x) = \int_D \kappa(x - y)\, v(y)\, dy, \tag{3}$$

which in Fourier space becomes a pointwise multiplication:

$$(\mathcal{K}v)(x) = \mathcal{F}^{-1}\big(R_\phi \cdot \mathcal{F}(v)\big)(x), \tag{4}$$

where R_ϕ is a tensor of learnable complex weights. The FNO implements this as a layer that (i) applies a 2D FFT, (ii) truncates to the lowest k Fourier modes, (iii) multiplies by learned complex weights, and (iv) applies the inverse FFT. A residual 1×1 convolution is added in parallel to capture local structure.

Because the operation is defined in frequency space, it is resolution-agnostic: the same learned weights can be applied to inputs at any spatial resolution.

2.3 U-Net

The U-Net [2] is a convolutional encoder–decoder architecture. An encoder successively halves the spatial resolution while doubling the channel count; a symmetric decoder upsamples back to the original resolution. Skip connections concatenate encoder feature maps into the corresponding decoder stages, preserving fine spatial detail. All operations are local spatial convolutions, so U-Net has no intrinsic mechanism for generalizing across resolutions.

3 Dataset

We use the `nsforcing_128` dataset from the neuraloperator Zenodo archive [3] (doi:10.5281/zenodo.12825163). The dataset consists of 12,000 pairs (x, y) where each is a 128×128 vorticity field: $x = \omega(t)$ and $y = \omega(t + \Delta t)$. We use 10,000 pairs for training and 2,000 for evaluation. Data is stored as PyTorch `.pt` tensors.

Input vorticity fields have low amplitude (std $\approx$ 0.15, range $[-0.76, 0.82]$), while target fields are more energetic (std ≈ 0.71, range $[-3.49, 3.93]$), reflecting the turbulent energy cascade over one timestep. The per-sample standard deviations are nearly linearly correlated, confirming the dataset is internally consistent. Figure 1 shows representative input–target pairs.

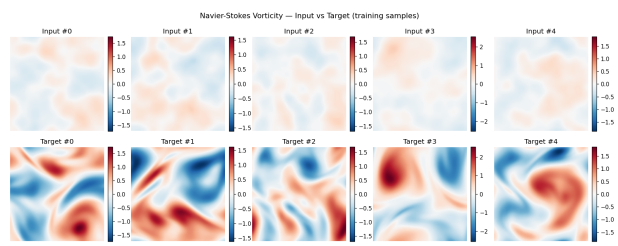

Figure 1: Five input–target vorticity pairs from the training set. Top row: input $\omega(t)$ (smooth, low amplitude). Bottom row: target $\omega(t + 1)$ (more turbulent, wider dynamic range). Color scale: RdBu_r, red = positive/clockwise vorticity, blue = negative/counter-clockwise.

4 Methods

4.1 FNO Architecture

Our FNO2d implementation follows Li et al. [1] closely. The input vorticity field (1 channel) is augmented with a 2D coordinate grid (gx, gy) to

produce a 3-channel input, then lifted to a 64-channel feature map via a learned 1×1 convolution. Four Fourier blocks follow, each performing a spectral convolution (truncated to 12 Fourier modes per spatial dimension) summed with a bypass 1×1 convolution, followed by GELU activation. The output is projected from 64 channels to 128, then to 1, via two further 1×1 convolutions with a GELU in between. Total parameters: **4.7M**.

4.2 U-Net Architecture

Our U-Net uses a 4-level encoder–decoder structure. The encoder doubles channels at each level (64→128→256→512) while halving spatial resolution via 2×2 max-pooling. The decoder uses bilinear upsampling and concatenates the corresponding encoder feature map at each level. Each block applies two 3×3 convolutions with BatchNorm and ReLU. Total parameters: **31M**.

4.3 Training

Both models are trained for 100 epochs on an NVIDIA GeForce RTX 2060 (6 GB VRAM) with the following settings:

- Loss: relative L^2 error, $\mathcal{L} = \|\hat{u} - u\|_2 / \|u\|_2$
- Optimizer: Adam, initial learning rate 10^{-3}
- LR schedule: cosine annealing to 10^{-5}
- Batch size: 20

Checkpoints are saved whenever validation loss improves. Training is logged with MLflow.

5 Results

5.1 Standard Evaluation

Table 1 reports test-set relative L^2 error, throughput, and latency for both models. FNO achieves roughly 3× lower error than U-Net while using 6.6× fewer parameters and running 2.2× faster.

Table 1: Standard evaluation on the 128×128 test set.

Model	Params	Rel-L^2↓	Throughput↑ (samp/s)	Latency↓ (ms/samp)
FNO	4.7M	**0.0348**	**499.2**	**2.003**
U-Net	31M	0.1029	227.0	4.406

Figure 2 shows side-by-side predictions from both models on six test samples spanning the range of FNO error (best to median to worst). FNO predictions closely match the ground truth in both large-scale structure and fine vortex detail. U-Net captures the broad structure but misses fine-grained vorticity features and exhibits higher pointwise error throughout.

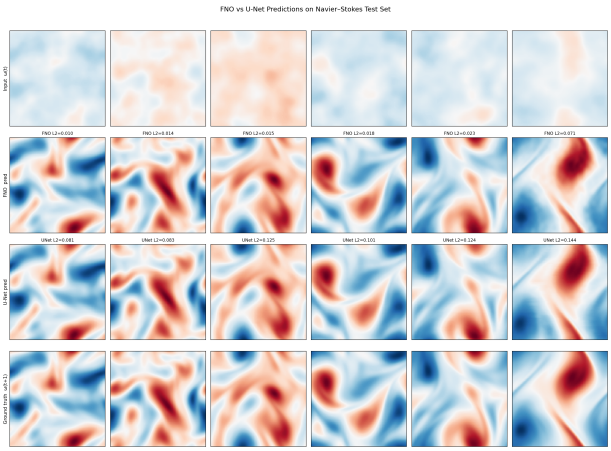

Figure 2: Six test samples ordered from lowest to highest FNO error. Rows: input $\omega(t)$ / FNO prediction / U-Net prediction / ground truth $\omega(t+1)$. Per-sample relative L^2 scores annotated per column.

Figure 3 shows absolute error maps $|\hat{u} - u|$ for both models. FNO errors are small and diffuse; U-Net errors are concentrated at vortex cores and high-gradient regions, consistent with difficulty learning fine-scale turbulent features.

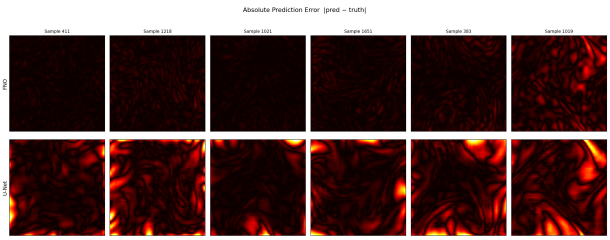

Figure 3: Absolute prediction error $|\hat{u} - u|$ for FNO (top) and U-Net (bottom) on the same six samples as Figure 2. Color scale: hot (black = zero, white = maximum). U-Net errors are systematically larger and concentrated at vortex boundaries.

Figure 4 shows the per-sample relative L^2 distribution across the full 2,000-sample test set. FNO's distribution is narrow and concentrated near 0.035; U-Net's is broad and shifted to the right, indicating consistently higher error with more variability.

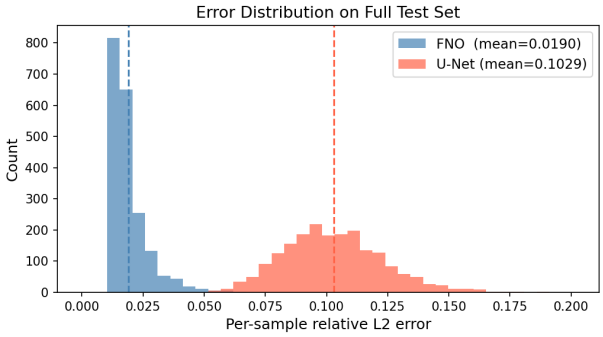

Figure 4: Distribution of per-sample relative L^2 error over the full test set. FNO (blue, mean = 0.035) is tightly concentrated; U-Net (red, mean = 0.103) is wider and skewed toward higher error.

5.2 Zero-Shot Super-Resolution

Table 2 and Figure 5 show results when both models—trained only at 128×128—are evaluated at 64×64 and 256×256 without any fine-tuning. Ground truth at each resolution is obtained by bilinearly resampling the 128 test targets.

Table 2: Zero-shot super-resolution: relative L^2 at unseen resolutions.

Resolution	FNO Rel-L^2	U-Net Rel-L^2
64×64	**0.0185**	0.4764
128×128 (train)	**0.0190**	0.1029
256×256	**0.0188**	0.5941

FNO's error is essentially flat across all three resolutions, confirming that the Fourier layers learn a resolution-invariant operator. U-Net's error at 64×64 and 256×256 is 4.6–5.8× higher than at its training resolution, demonstrating that fixed spatial kernels do not generalize across grid spacings.

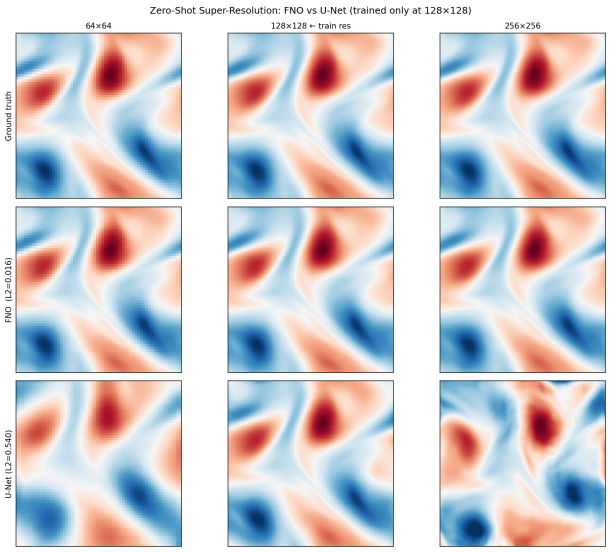

Figure 5: Zero-shot super-resolution on a single test sample. Columns: 64×64, 128×128 (train resolution), 256×256. Rows: ground truth / FNO prediction / U-Net prediction. FNO maintains consistent visual quality at all resolutions. U-Net produces smooth but blurry outputs at 64×64 and introduces artifacts at 256×256.

6 Discussion

Why FNO outperforms U-Net. The performance gap stems from an inductive bias alignment. The Navier–Stokes operator is global in nature: a vortex anywhere in the domain can in-

fluence the flow everywhere else through the pressure field. FNO captures this via global Fourier modes; U-Net's local 3×3 kernels must compose many layers to propagate information across the domain. Additionally, the FFT-based parameterization of FNO directly represents the low-frequency energy content of the vorticity field, which dominates NS dynamics.

Resolution invariance. FNO's flat super-resolution curve is a direct consequence of Fourier convolutions operating in frequency space: the learned weights multiply specific Fourier coefficients, which carry the same physical meaning regardless of how many spatial grid points discretize the domain. The U-Net's degradation at unseen resolutions is expected: 3×3 kernels at 128×128 correspond to a specific physical length scale, which changes when the grid spacing changes.

Parameter efficiency. Despite having $6.6\times$ fewer parameters, FNO outperforms U-Net substantially. This is consistent with the general finding that architecture-level inductive biases can substitute for raw model capacity when the biases are well-matched to the problem structure.

Limitations. Several limitations should be noted. First, both models are trained and evaluated at a single fixed viscosity $\nu = 10^{-4}$; generalization across viscosities or forcing functions is not tested. Second, we evaluate only single-step prediction; autoregressive rollout over many timesteps is likely to accumulate error. Third, the super-resolution ground truth is obtained by bilinear downsampling of 128×128 data, not from independent high-resolution simulations, so the 256×256 evaluation is a proxy.

7 Conclusion

We trained and evaluated a Fourier Neural Operator and a U-Net on 2D Kolmogorov-forced Navier–Stokes time-stepping. FNO achieves $3\times$ lower relative L^2 error with $6.6\times$ fewer parameters and $2.2\times$ higher inference throughput. In zero-shot super-resolution, FNO's error remains flat while U-Net's degrades by up to $5.8\times$.

These results support the conclusion that neural operators, which encode physical structure—specifically, the frequency-space nature of PDE operators—directly into the architecture, outperform generic convolutional baselines on PDE surrogate modelling tasks. The FNO framework represents a principled step toward fast, data-efficient neural simulators that can generalize across discretizations.

References

[1] Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2021). Fourier Neural Operator for Parametric Partial Differential Equations. *International Conference on Learning Representations (ICLR).* arXiv:2010.08895.

[2] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241.

[3] neuraloperator. (2024). NS Forcing Dataset 128. Zenodo. doi:10.5281/zenodo.12825163.